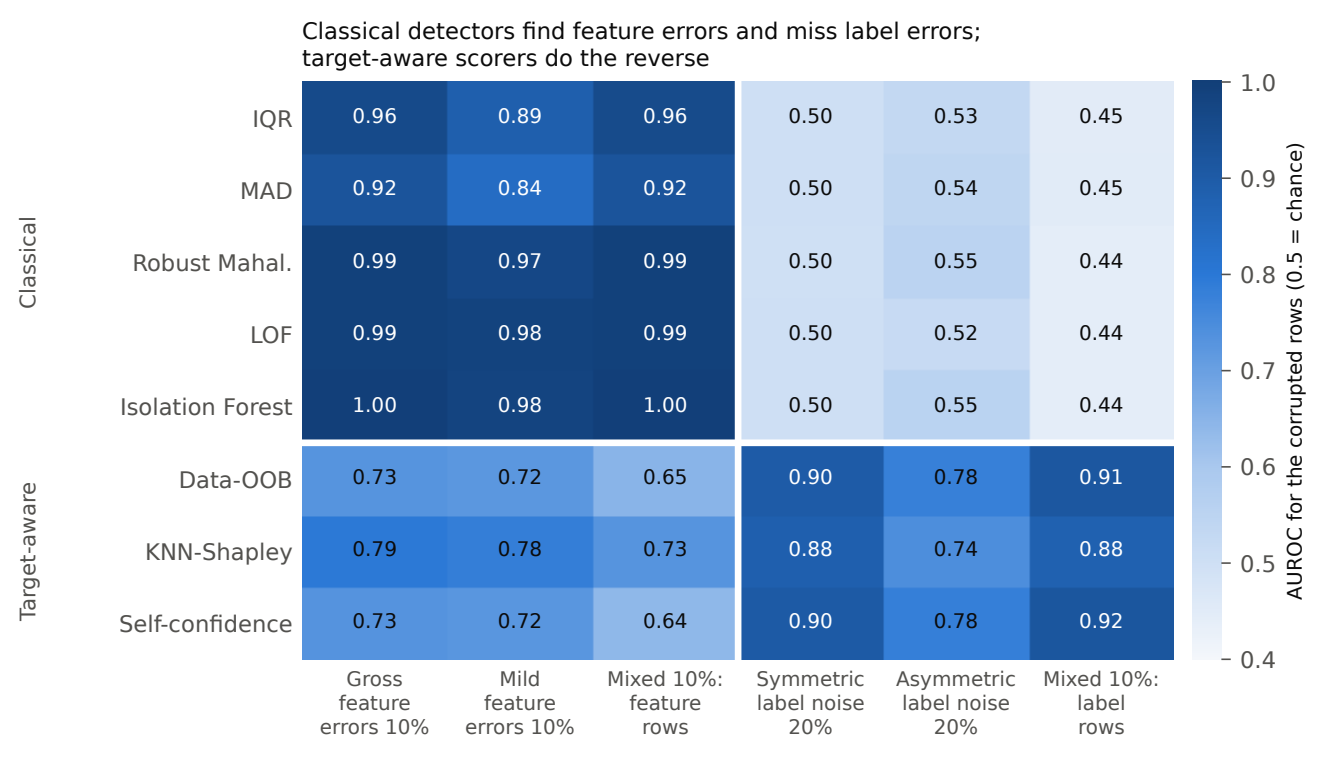
Classical detectors find feature errors and miss label errors;
target-aware scorers do the reverse
Classical
Target-aware
IQR
MAD
Robust Mahal.
LOF
Isolation Forest
Data-OOB
KNN-Shapley
Self-confidence
0.96 0.89 0.96 0.50 0.53 0.45
0.92 0.84 0.92 0.50 0.54 0.45
0.99 0.97 0.99 0.50 0.55 0.44
0.99 0.98 0.99 0.50 0.52 0.44
1.00 0.98 1.00 0.50 0.55 0.44
0.73 0.72 0.65 0.90 0.78 0.91
0.79 0.78 0.73 0.88 0.74 0.88
0.73 0.72 0.64 0.90 0.78 0.92
Gross feature errors 10%
Mild feature errors 10%
Mixed 10%: feature rows
Symmetric label noise 20%
Asymmetric label noise 20%
Mixed 10%: label rows
AUROC for the corrupted rows (0.5 = chance)
1.0
0.9
0.8
0.7
0.6
0.5
0.4